Research Progress Report 3

Site Dependent Anisotropy in the Heisenberg Model

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1. Research Direction

Following the previous XY and Heisenberg simulations, the next step was to test what happens when strong magnetic anisotropy is present on only a fraction of the lattice sites. This is intended as a simplified model for reducing the amount of strongly anisotropic material while retaining enough anisotropy to produce a useful coercive field.

The main questions in this set of simulations are how the coercive field changes as the fraction of strongly anisotropic sites is reduced, and whether the spatial arrangement of those sites matters. I therefore compared regular periodic arrangements with randomly distributed strong anisotropy sites.

2. Site Dependent Anisotropy Model

The Heisenberg model is the same as in the previous report, except that the anisotropy strength can now vary from site to site. The Hamiltonian can be written as:

$$E = -J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j - \sum_i D_i (S_i^x)^2 - H \sum_i S_i^x$$

Each lattice site is assigned a local anisotropy value Di. Strong anisotropy sites use Dmax, while the remaining sites use Dmin. For the current simulations, Dmax = 10 and Dmin = 0.1, giving a factor of 100 between the two anisotropy strengths.

For random configurations, each site is assigned Dmax independently with probability p and Dmin with probability 1-p. The resulting anisotropy map is generated once at the beginning of a hysteresis calculation and remains fixed throughout the full magnetic field sweep.

For the periodic cases, the strong anisotropy sites are distributed in regular patterns such as a checkerboard at p = 0.5 and evenly spaced sublattices at lower concentrations with adjustable spacings.

3. Simulation Conditions

The comparisons were performed on a 10x10 Heisenberg lattice with J = 1 and T = 0.25. The external field was swept from H = +4 to H = -4 and back to H = +4 using a field step dH = 0.2. At each field value, 100 Monte Carlo sweeps were performed before recording the magnetization. The coercive field was estimated from the M = 0 crossings of the descending and ascending branches and averaged between the two sides.

Each condition was repeated 20 times so a total of 20 hysteresis curves were produced per condition. To save space, I have disabled python from producing a plot for each individual run and instead recorded only the coercive values.

For periodic arrangements, the spatial anisotropy pattern is the same in each run and the spread mainly reflects Monte Carlo randomness.

For random arrangements, the spread includes both Monte Carlo randomness and differences between independently generated anisotropy configurations.

4. Coercive Field as the Strong Anisotropy Percentage is Reduced

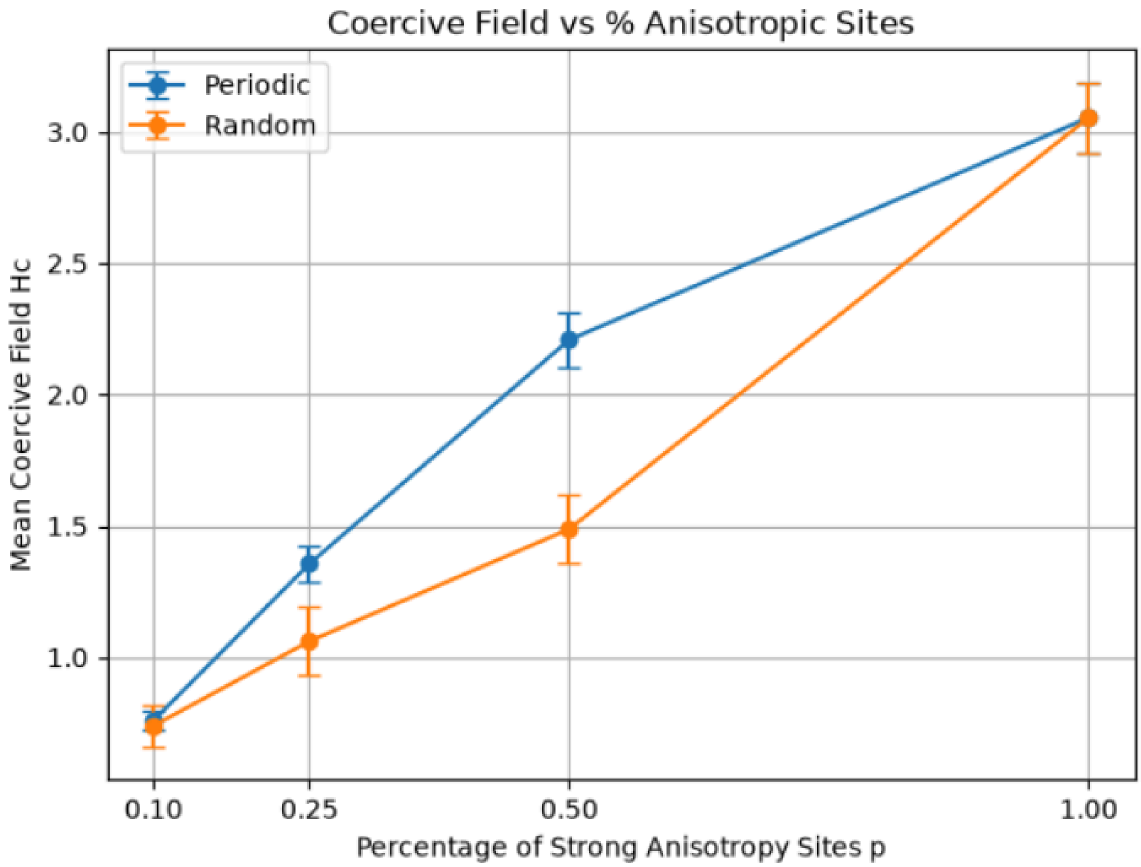

Figure 1: Mean coercive field as a function of the fraction p of strong anisotropy sites. Error bars show the standard deviation over 20 runs.

The coercive field decreases as the fraction of strongly anisotropic sites is reduced.

At p = 1, this arrangement represents where all sites have strong anisotropy, the mean Hc is 3.054.

At p = 0.5, the checkerboard arrangement has a mean Hc of 2.209, while the random arrangement gives 1.488.

At p = 0.25, At p = 0.25, the periodic and random cases decrease further to 1.356 and 1.061 respectively.

At p = 0.10, both arrangements produce a much smaller coercive field of about 0.75.

Strong-D fraction	Arrangement	Mean Hc	STD
1.00	all sites	3.054	0.132
0.50	checkerboard	2.209	0.106
0.50	random	1.488	0.128
0.25	periodic	1.356	0.068
0.25	random	1.061	0.129
0.10	periodic	0.760	0.039
0.10	random	0.740	0.081

Table 1: Mean coercive field and standard deviation for the tested strong anisotropy percentages and spatial arrangements.

5. Effect of Spatial Arrangement

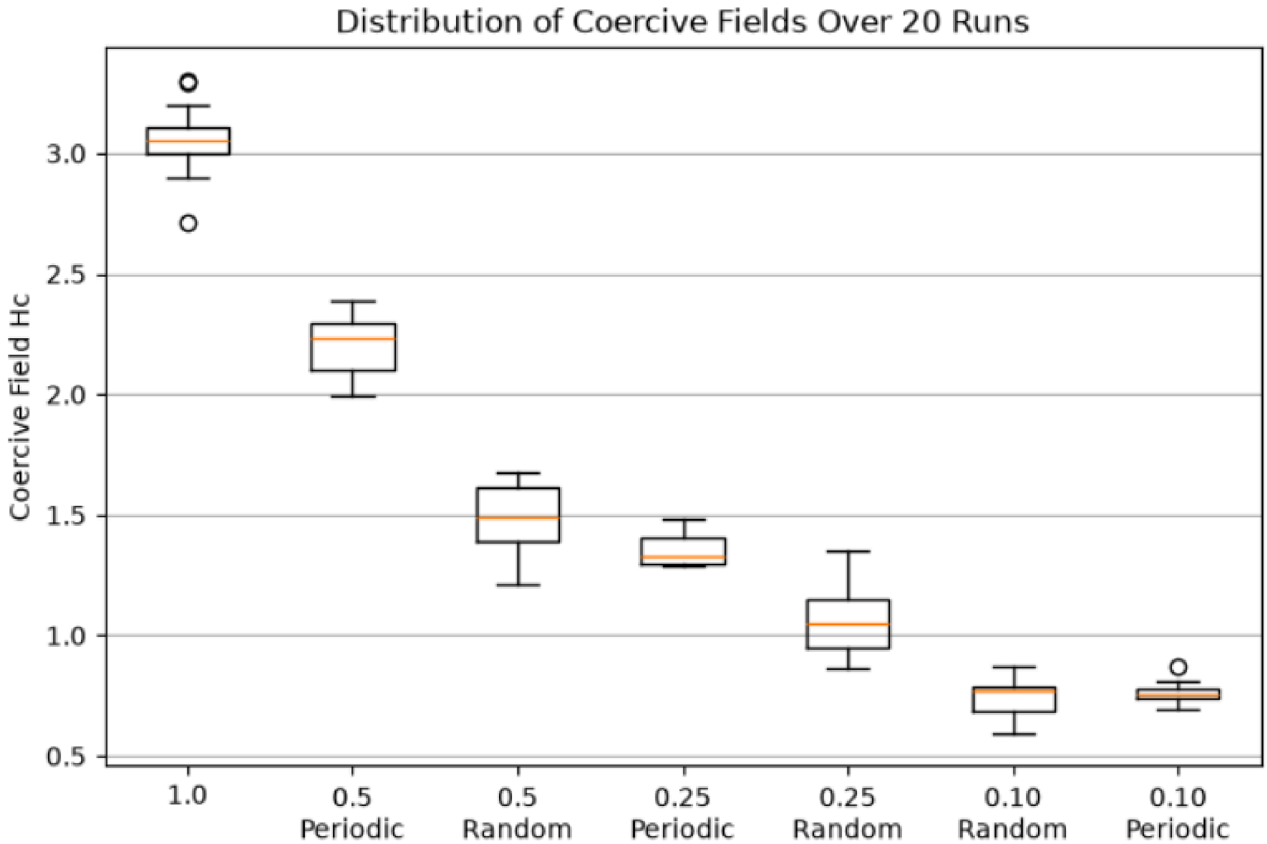

Figure 2: Distribution of coercive-field values over the repeated simulations for periodic and random anisotropy configurations.

The spatial arrangement has a noticeable effect at intermediate concentrations.

At p = 0.5, the checkerboard configuration produces a substantially larger coercive field than the random configuration.

A similar but smaller difference can be seen at p = 0.25.

By p = 0.10, the periodic and random mean coercive fields are nearly the same.

The random configurations also generally show a larger standard deviation than the periodic cases. This is expected because the random simulations contain an additional source of variation where each run has a different arrangement of strong and weak anisotropy sites. The results suggest that at intermediate concentrations, distributing the strong anisotropy sites more uniformly may stabilize the magnetized state more effectively than allowing them to form random clusters and gaps.

The main trend is that reducing the concentration of strongly anisotropic sites progressively reduces the coercive field, but a partial concentration can still retain a substantial fraction of the full anisotropy coercivity. The results also indicate that the spatial distribution of anisotropy can matter in addition to the total percentage of strong anisotropy sites.

6. Potential Future Implementations

The current model treats the main source of material variation as the local anisotropy strength Di. A natural next step would be to generalize this into a more complete material map in which different lattice sites can have several different magnetic properties. For example, each site could be assigned its own easy-axis direction rather than having every anisotropy term favor the global x direction. The anisotropy contribution could then be written in the form $-D_i(S_i \cdot n_i)^2$, where n_i is the local easy-axis direction. This could be used to represent imperfectly aligned grains by assigning a common easy axis to groups of neighboring sites and varying the orientation between grains.

Another extension would be to allow the exchange interaction J to depend on the materials of the two neighboring sites. Instead of using the same J for every pair, the model could distinguish between interactions within the strongly anisotropic material, within the weakly anisotropic material, and across the interface between the two. This would introduce parameters such as J_{AA}, J_{BB}, and J_{AB}. The interface coupling could then be varied to study how strongly the high anisotropy regions must be coupled to the surrounding material in order to maintain a large coercive field.

The current random anisotropy model could also be extended to study spatial structure more systematically. Instead of assigning strong anisotropy sites independently, they could be arranged into clusters, grains, or separated by controlled distances while keeping the total fraction p fixed. This would help determine whether coercivity depends mainly on the amount of strongly anisotropic material or also on how that material is distributed. The difference already observed between periodic and random arrangements suggests that spatial distribution may be important.

Other possible improvements include assigning different magnetic moment strengths to the two materials, introducing weaker exchange coupling at grain boundaries, allowing a distribution of anisotropy strengths rather than only two fixed values, and applying the magnetic field at different angles relative to the easy axes.

Other extensions could include a 3D lattice or dipole-dipole interactions, although these would substantially increase the computational time. These additions would gradually move the simulation from a simple site dependent anisotropy model toward a more general model of a heterogeneous magnetic material.